

# DSA ki Baithak



**CODE  
BAITHAK**

BY NAMAN BHAIYA



# # What is Data Type ?

[Container = Variable]

A data type defines :

- \* Type of data to be stored
- \* Size of memory to be allocated
- \* Range of values

[Java is Strongly-typed language]

Ex: int, float, char, boolean,  
String, Array, etc

# # Types of Data Types in Java

In Java, there are two-types of Data Types:

① Primitive Data Types - store simple values directly  
[which cannot be further broken down] in the memory

② Non-Primitive (Reference) Data Types -  
store memory references  
to objects

# # Primitive Data Types

There are 8 - Primitive data types in Java -

## Integer

- \* byte
- \* short
- \* int
- \* long

## Decimal

- \* float
- \* double

## Character

- \* char

## True/False

- \* boolean

# # Integer Data Types

[stores the integer type  
of values]

Data Type

Size

[1 byte = 8 bits]

byte

1 byte

short

2 bytes

int

4 bytes

long

8 bytes

['bit' is the smallest  
unit of memory]

[default integer → int  
literal]

# # Decimal Data Types

[stores the decimal  
type of values]

Data Type

Size

float

4 bytes

double

8 bytes

[default decimal → double  
literal]

# # Character & Boolean Data Types

[ char → stores the single character  
boolean → stores either true or false ]

Data Type

char

boolean

Size

2 bytes

JVM dependent

[ char uses Unicode encoding ]



\* In Java, char internally does not store the symbol,  
it actually stores the numeric code (Unicode value).

internally character = number

# ASCII Concept

old - limited to English only

[ Both concepts map  
characters to the numbers ]

# Unicode Concept

new - supports all languages

\* First 128 characters of ASCII & Unicode  
are same.

Characters

'A' - 'Z'

'a' - 'z'

'0' - '9'

ASCII values

65 - 90

97 - 122

48 - 57